

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Homework 6: Fitting Data and Scientific Data Presentation
Due Monday, October 12, 2026, 2:30 pm

Remember, at all points, to explain what you do in your log and keep appropriate notes in your code!

1. **(5 points)** Use your terminal to make a new folder `hw7_<username>` under your `ast4762/` homework folder. For this homework your main homework file is a Python file named `hw7_<username>.py` (or `hw7_<username>.ipynb` but remember to save it as a `.py` file before submission). Save it in your homework folder. Remember to commit your files and push to GitHub. Start the file with the usual header.

2. **(40 points)** As part of your final project, you will need to write a scientific paper. As we discussed in class, for this you will need to write an introduction that presents the topic you study and what your contribution to it is. Familiarize yourself with the literature about the target of the AST4762 project (everyone) by reading the following papers about the dataset that you will use:

- a) <https://arxiv.org/abs/0705.0993>
- b) <https://arxiv.org/abs/0802.0845>

Then, learn some more-recent information about the target by reading at least two papers like these:

- c) <https://arxiv.org/abs/1307.3239>
- d) <https://arxiv.org/abs/2410.22398>

For each paper, write a 1-2 paragraph overview of the key points you learned from each paper about the target. **You may not use AI for this.**

3. **(50 points total)** You observed an exoplanet transit and got the data given in file `hw7_synthetic_transit_lightcurve.csv`.

a) **(10 points)** Read the data and plot the observations in a publication-ready plot.

b) **(30 points total)** One can retrieve the radius of the planet from the depth of the transit using:

$$\frac{\Delta F}{F} = \left(\frac{R_p}{R_*} \right)^2,$$

where R_p is the planetary radius, R_* the stellar radius, and $\frac{\Delta F}{F}$ is the fractional decrease of flux during the transit. Here, we will use a simple box-window function to fit the data.

- i) **(8 points)** Make a box (rectangular) window function. The function should take as input an array for the x values (this will be the observational time), the location of the center of the window, and the width and the height of the function.

- ii) **(8 points)** Use `curvefit()` to fit the transit data with the window function. Can you believe the fit? What should you do if you can't?
- iii) **(4 points)** What are the best-fit window-function properties? What is the exoplanet radius (in $R_{\odot}$), if you know that the stellar radius is $1.14 R_{\odot}$?
- iv) **(10 points)** Make a publication-ready plot showing the data and the best-fit model.

c) **(10 points)** The stellar radius is measured either with asteroseismology or fitting with stellar evolution models, so there is always an error associated with it. Assume that the error in the stellar radius is $0.01 R_{\odot}$ and that your fit has an error in the fractional depth of 0.002 (regardless of what the fit says the error was). What is the error in the planetary radius?

4. **(5 points)** Test that your code works on Stokes and submit your homework. Follow the procedures in the final problem of earlier homeworks.